# K.A.W.H.I – Kinetic Adaptation via Wasserstein Heuristics and Identity

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**Abstract**

The rise of spatiotemporal tracking data has shifted basketball analytics from discrete events to continuous, team-level dynamics. While current simulation methods often rely on imitation learning, we propose a physics-based approach that models a basketball defense as a continuous fluid or "active matter" system rather than independent agents. Using 2015-16 NBA SportVU tracking data, we formulate defensive transitions as a Wasserstein Gradient Flow utilizing the Jordan-Kinderlehrer-Otto (JKO) scheme. We define a free energy functional comprising collective team potentials—such as cohesion, occupancy, and ball pressure—alongside geometric constraints like anisotropic offender attraction and basket gravity. The JKO scheme balances this potential energy minimization against the kinematic cost of optimal transport, approximated via Sinkhorn divergence, to ensure physically realistic player trajectories. Hyperparameters for these potentials are tuned via Bayesian Optimization to minimize pressure and smoothness losses. While steep potential gradients can occasionally induce simulated speeds exceeding human constraints, the Wasserstein framework inherently regularizes motion. Ultimately, this demonstrates the viability of optimal transport and thermodynamic principles for continuous spatiotemporal sports modeling.

Website: https://takafumim.github.io/Wasserstein-Gradient-Flows/
Code: https://github.com/cadenpascual/NBA-defensive-dna

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# 1 Introduction

Historically, the evolution of basketball analytics has been dictated by data availability. Up until the 1990s, teams used discrete box score statistics, points, rebounds, assists, to quantify player performance. The late 2000s, labeled the "Moneyball" era, utilized the rise of computing to introduce advanced metrics and efficiency ratings, using more in-depth statistics to value players who may not fill the stat sheet.

Today, we have entered the age of spatiotemporal tracking data. Through the use of optical-tracking cameras, teams are able to turn player micro-movements into data to inform decision making. With this tracking data, a team is no longer reduced to a collection of individual players, but can be viewed as a unified entity. This "team-as-entity" approach gives us macro-motion properties that are distinct from, and often more predictive than the sum of individual actions by the players.

While many current models rely on Imitation Learning, training neural networks to mimic player movements, we propose a physics-based alternative. We treat the defense as an "Active Matter" system that flows across the court, minimizing a structured energy functional. In mathematics, the Jordan-Kinderlehrer-Otto (JKO) scheme proved that such collective motion and diffusion can be modeled as a gradient descent in Wasserstein space. Just as a physical object rolls down a hill to minimize potential energy, a probability distribution flows to minimize its "Free Energy."

By treating the five player defensive unit as a distribution of mass, this paper addresses a fundamental question: Can the transition of a basketball defense be mathematically modeled as a Wasserstein Gradient Flow?

# 2 Data Pipeline

NBA tracking data is known for being difficult to work with due to its size and complex structure. Before modeling collective team dynamics, we first had to build a custom big data pipeline. The system allowed us to structure the raw data, letting us run high-speed simulations for our physics model.

## 2.1 Raw Tracking Data Characteristics

Our primary dataset consists of spatiotemporal SportVU tracking data, spanning over 600 games and 90,000 shots from the 2015-2016 NBA season. The SportVU optical tracking system records the $(x, y)$ coordinate positions of all ten players on the court, alongside the $(x, y, z)$ coordinates of the ball, at a sampling rate of 25 frames per second.

Within the raw dataset, individual games were stored as highly compressed `.7z` archives. Our initial processing step required extracting these archives into raw `.json` files, which were structured as massive, nested arrays grouped into "events" (plays) and "moments"

(frames). While this structure captured the continuous micro-movements of the game, these frames were unlabeled, providing spatial geometry but lacking context. Furthermore, initial data exploration revealed that loading and storing entire games of nested JSON arrays into memory was computationally infeasible for continuous energy modeling.

### 2.2 Play-by-Play and Shot Chart Integration

Because raw tracking data strictly records geometry without recording official basketball outcomes (e.g., whether a specific play resulted in a made basket, turnover, etc.), it was necessary to integrate an external dataset that filled in where tracking data was incomplete. To achieve this, we utilized the `nba_api` to fetch complete play-by-play and shot chart logs for the entire 2015-2016 season.

This supplementary dataset provided contextual metadata for every single field-goal attempt in the season. It included the location of the shot, the shooter's identity, and a binary make-or-miss label. By acquiring this discrete event data, we established the foundation to build our Expected Field Goal ($xFG$) and Instantaneous Shot Threat ($IST$) representations, which inherently require a target variable ($shot_made$) to measure the empirical impact of a defense's spatial positioning.

## 3 Data Alignment

In order to model defense as a continuous system, we segmented the raw data into isolated possessions ending in a shot. Because the tracking data lacks event outcomes, and the Play-by-Play (PBP) data lacks context prior to the shot, we developed a custom alignment protocol to bridge this information gap.

### 3.1 Heuristic Event Merging

A primary challenge in data integration was that the SportVU feed periodically drops possessions. As a result, the discrete `EVENT_ID` values in the SportVU database are inconsistent with the official identifiers in the NBA API. This lack of relational consistency made standard tabular joining impossible. To resolve this, for every event in the tracking database, we calculated the absolute start and end time using the `game_id`, `period`, and `game_clock`. These intervals were then cross-referenced against the timestamps of the NBA API shot logs. We enforced a strict 75-frame (3-second) minimum threshold for each possession to provide sufficient context for momentum calculations and to stabilize the initial defensive topology. Finally, we algorithmically identified and removed redundant overlapping events caused by tracking system resets to ensure data integrity.

## 3.2 Release Frame Synchronization

Initial data verification revealed a systematic temporal lag (typically 3-4 second delay) in PBP shot timestamps relative to the physical release of the basketball. To ensure our models evaluate defensive positioning at the moment of peak threat, we developed a trajectory-based alignment protocol.

By backtracking from the delayed timestamp to the frame where the ball begins its vertical ascent, our pipeline determines the "True Release Frame." This synchronization ensures that all spatial and threat metrics, including $IST$ and $xFG$, reflect the exact defensive configuration at the moment the shot is taken.

Building on threshold established in section 3.1, we extracted a dynamic temporal window for each valid possession beginning 75 frames (3 seconds) prior to the True Release Frame. This sequence also extends past the delayed PBP timestamp to capture the visual outcome of the shot. To optimize computation, these sequences were saved as compressed `.parquet` files. A `local_release_idx` bookmark was used to allow the JKO engine to only look at the 75 pre-shot frames.

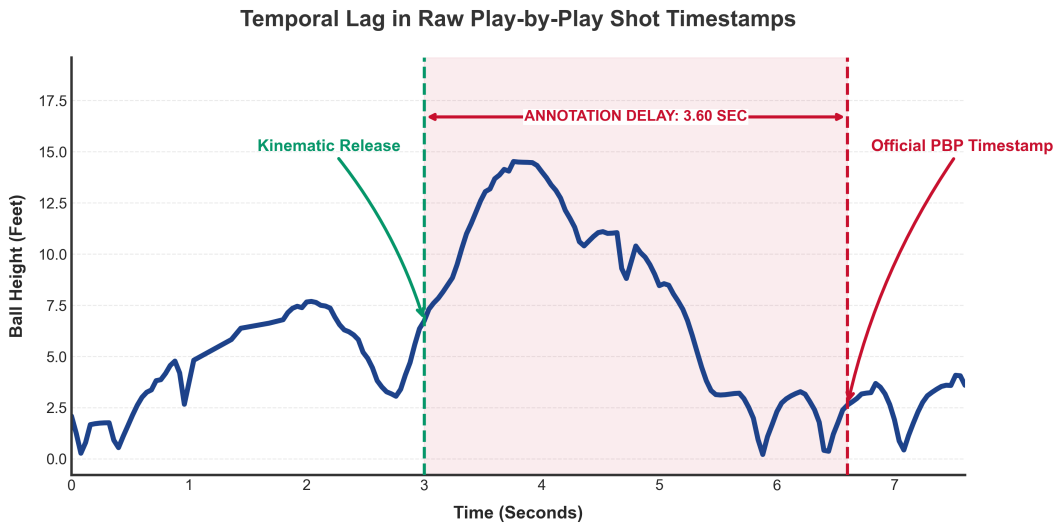



Figure 1: **Annotation Delay in Play-by-Play Data.** A representative 3-point jump shot demonstrating the temporal lag present in raw NBA event logs. The red dashed line indicates the delayed Official PBP Timestamp, while the green dashed line represents our algorithmically derived release frame. On this specific possession, the human annotation was delayed by 3.6 seconds, highlighting the need for algorithmic frame correction.

# 4 Representation Learning

To transform raw tracking coordinates into a physics engine, we model the court as as a continuous energy surface. Through this framework, offensive players exert a spatial force, proportional to their scoring presence, that the defensive unit must flow to neutralize.

## 4.1 Calibrated xFG and Bayesian Player Priors

To establish a baseline for the scoring potential of a given shot, we first trained a Base Expected Field Goal (`xFG_base`) model. We used a Logistic Regression classifier on spatial features, including `shot_dist`, `angle`, `is_three`, and `is_corner`. To ensure temporal integrity, we split the season chronologically:

- **Training:** Prior to 2016-01-01 (494 games)
- **Validation:** 2016-01-01 to 2016-02-15 (320 games)
- **Test:** Post 2016-02-15 (416 games)

While `xFG_base` captures the difficulty of a shot's geometry, it ignores the identity of the shooter. To account for individual skill, we introduced a **Bayesian Player Prior**. This prior was calculated independently for each shot-zone to reflect player-specific spatial tendencies. To prevent low-volume shooters from skewing the distribution, we applied Bayesian shrinkage, pulling their raw shooting rate toward the league average using an empirical weight of $\alpha = 300$:

$$\text{Player Prior} = \frac{\text{makes} + \alpha \cdot \text{league_rate}}{\text{attempts} + \alpha} \tag{1}$$

Furthermore, any player with fewer than 30 attempts in a shot zone was defaulted to the league mean. This spatial prior, alongside `is_three` and `xFG_base`, was integrated into a secondary regression model.

Finally, in order to align predicted probabilities with observed frequencies in the real data, we calibrated the second model using isotonic regression on the validation set. The model demonstrated strong predictive stability across the temporal split, achieving a test-set Brier score of 0.229 (validation: 0.230) and log-loss of 0.649 (validation: 0.651). This pipeline yields our final `xFG_offense` metric, a mathematically sound representation for shot value.

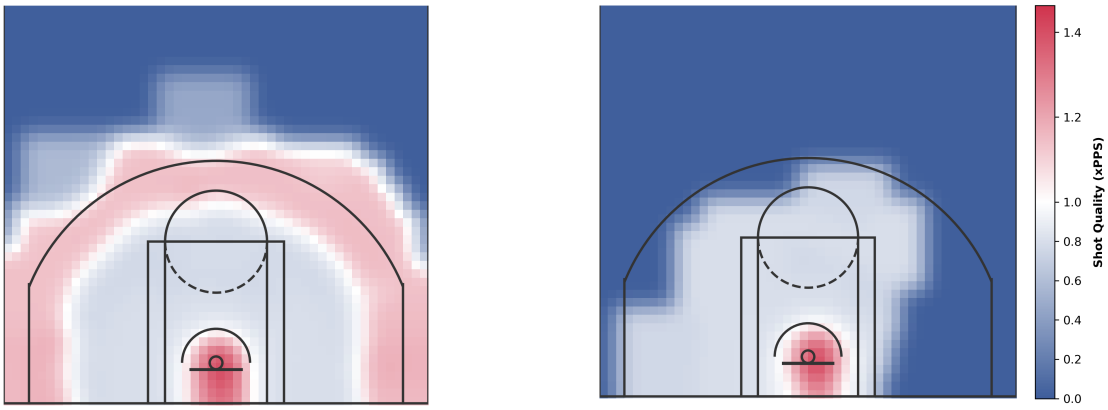
## 4.2 Spatial Quality Maps ($Q$) and xPPS

Because the JKO scheme requires continuous spatial gradients rather than discrete data points, we translated `xFG_offense` into 2D expected value surfaces. First, we converted the expected Field Goal probability into Expected Points Per Shot (`xPPS_offense`) by multiplying the probability by the fundamental shot value (2 or 3 points). This ensures that our model takes into account the threat of a 3-point shot, even if the `xFG_offense` was lower.

Next, we discretized the offensive half-court into a $1 \times 1$ ft coordinate grid. For each player, we aggregated their `xPPS_offense` values into a spatial matrix. Because the JKO framework requires smooth, differentiable potentials, we applied a 2D Gaussian filter ($\sigma = 1.25$) to transform our grid into a continuous field. This allows the physics engine to differentiate between distinct spatial threat profiles, as visualized in Figure 2.

To preserve statistical validity, individual $Q$ maps were only generated for players with $>$ 100 shot attempts; others were assigned to a universal "League Average" map, aggregated

from the global dataset. To optimize our pipeline, player maps were saved as highly compressed `.npz` tensor arrays, allowing the physics engine to execute $O(1)$ queries to retrieve the spatial threat of any player at any given coordinate.




(a) Kawhi Leonard: Versatile Threat

(b) Rudy Gobert: Rim-Centric Threat

Figure 2: **Spatial Quality Maps ($Q$).** A comparison of smoothed expected points per shot ($xPPS$) surfaces for two distinct player archetypes. **(a)** Kawhi Leonard demonstrates a broad, all-around threat surface extending beyond the three-point arc. **(b)** Rudy Gobert exhibits an extreme, localized threat strictly confined to the restricted area.

## 4.3 Instantaneous Shot Threat (IST)

Simulating a realistic defense requires the ability to prioritize offensive threats dynamically. While standard physics models treat offensive players as uniform geometric points, in real life, defenders react to specific players based on their shooting strengths. We developed the Instantaneous Shot Threat (IST) to bridge this gap, allowing the JKO engine to quantify real-time scoring effectiveness, so defenders "sag" off weak perimeter threats and close out aggressively on elite scorers.

We first established IST as a log-linear interaction between Shot Quality ($Q$), Openness ($O$), and Shootability kinematics ($S$):

$$\log(\text{IST}) = \beta_0 + \beta_Q \log(Q) + \beta_O \log(O) + \beta_S \log(S)$$

Optimization via the **L-BFGS-B** algorithm revealed that coefficients ($\beta_S$) converged to zero, proving that shot quality and openness overwhelmingly dictate success. Furthermore, we identified a strict contest threshold of $d_0 = 4.80$ ft. The final simulated IST penalty for each offender $p_o \in P_O$ is calculated through the following steps:

1. **Ball Proximity Weight**: Scales the baseline threat based on the distance to the ball ($d_{\text{ball}}$) to account for off-ball players.

$$W_B = 0.4 + \frac{0.6}{1.0 + \exp(0.3(d_{\text{ball}} - 18.0))}$$

2. **Base Threat**: Fuses the spatial map with proximity, enforcing a minimum threat floor.

$$W_{\text{base}} = \max\left(Q(p_o)^{\gamma_Q} \cdot W_B, 0.35\right)$$

3. **Defensive Openness**: Translates the 4.80 ft empirical finding into a differentiable multiplier ($O$). We smooth the event horizon using a 6.0-foot spatial denominator and a continuous `clip` function, utilizing a `softmin` operator to identify the nearest defender while maintaining a stable gradient:
   - $d_{\text{closest}} = -\frac{1}{k_{\text{smooth}}} \ln\left(\sum_{p_d \in P_D} e^{-k_{\text{smooth}} \cdot \|p_d - p_o\|}\right)$
   - $O = \text{clip}\left(\frac{d_{\text{closest}}}{6.0}, 0.5, 1.5\right)$

The total energy penalty applied to the defensive system is the weighted sum of these individual threats:

$$E_{\text{IST}}(P_D, P_O, p_{\text{ball}}) = W_{\text{ist}} \cdot \sum_{p_o \in P_O} (W_{\text{base}} \cdot O^{\gamma_O}) \tag{2}$$

# 5 Methods

## 5.1 JKO Scheme

The Jordan-Kinderlehrer-Otto (JKO) scheme provides a powerful variational framework for interpreting and solving certain parabolic partial differential equations (PDEs) as gradient flows in the space of probability measures. Introduced by Jordan, Kinderlehrer and Otto (1998), it establishes a deep connection between optimal transport, thermodynamics, and diffusion processes. At its core, the JKO scheme formulates the time evolution of a probability density not as a traditional PDE, but as a sequence of optimization problems. It seeks to minimize a free energy functional, penalized by the cost of moving the probability mass, measured via the Wasserstein distance.

### 5.1.1 Mathematical Formulation

Let $\mathscr{P}_2(\mathbb{R}^d)$ denote the space of probability measures on $\mathbb{R}^d$ with finite second moments. Given a time step $\tau > 0$ and an initial probability density $\rho_0$, the JKO scheme constructs a sequence of densities $\{\rho_k\}_{k=1}^{\infty}$ where each $\rho_{k+1}$ is obtained by solving the following proximal-implicit Euler step:

$$\rho_{k+1} = \underset{\rho \in \mathscr{P}_2(\mathbb{R}^d)}{\text{argmin}} \left\{ \frac{1}{2\tau} W_2^2(\rho, \rho_k) + \mathscr{F}(\rho) \right\}$$

$W_2(\rho, \rho_k)$ is the 2-Wasserstein distance, representing the optimal transport cost to deform the density $\rho_k$ into $\rho$. $\mathscr{F}(\rho)$ is a free energy functional driving the system.The term $\frac{1}{2\tau} W_2^2(\rho, \rho_k)$ acts as an inertial penalty, restricting how far the distribution can move in a single time step $\tau$, analogous to a kinetic energy term.

This optimization reveals a fundamental physical competition: the system wants to immediately minimize its free energy $\mathscr{F}(\rho)$, but is constrained by the geometry of optimal transport, which resists instantaneous, infinite-velocity rearrangement of mass.

### 5.1.2 The Connection to Gradient Flows and PDEs

The true power of the JKO scheme lies in its continuous-time limit. As the time step $\tau \to 0$, the sequence of discrete measures $\rho_k$ converges to a curve $\rho_t$ that solves a continuous gradient flow equation in the Wasserstein space. A classic example is the recovery of the Fokker-Planck equation. Suppose the free energy functional consists of a potential energy term and an internal energy (entropy) term:

$$\mathscr{F}(\rho) = \int_{\mathbb{R}^d} V(x)\rho(x)dx + \beta^{-1} \int_{\mathbb{R}^d} \rho(x) \log \rho(x) dx$$

Where $V(x)$ is a confinement potential and $\beta$ is the inverse temperature. Under the JKO scheme, taking the limit $\tau \to 0$ yields the continuous time evolution governed by the continuity equation:

$$\frac{\partial \rho}{\partial t} = \nabla \cdot \left( \rho \nabla \left( \frac{\delta \mathscr{F}}{\delta \rho} \right) \right)$$

Substituting the variational derivative of the energy functional $\frac{\delta \mathscr{F}}{\delta \rho} = V(x) + \beta^{-1}(\log \rho + 1)$, we recover the linear Fokker-Planck equation:

$$\frac{\partial \rho}{\partial t} = \nabla \cdot (\rho \nabla V) + \beta^{-1} \Delta \rho$$

This demonstrates that diffusion (the Laplacian term) is geometrically equivalent to the gradient flow of the Shannon entropy with respect to the Wasserstein metric.

### 5.1.3 Application

We initially adopted the JKO framework because we were exploring a DFFT approach from Barron (2025) to model player dynamics. However, as we pivoted to using optimized Gaussian potentials to represent the ball and players, we realized the Wasserstein gradient flow provided by the JKO scheme solved several physical modeling issues that standard Euclidean gradient descent or Normal Gradient Flow algorithms such as Proximal Point Algorithm could not.

Standard gradient descent simply points in the direction of steepest descent. If a potential well suddenly appears far away, standard gradient descent might calculate an unrealistically massive step for a player to get there.

Normal Gradient Flow doesn't care about the spatial distance between the locations; it only cares about the change in value at each point.

In the JKO scheme, the $\frac{1}{2\tau}W_2^2(\rho, \rho_k)$ term acts as a fundamental kinematic penalty. The Wasserstein distance measures the exact cost of transporting mass from one area of the court to another. It intrinsically limits unnatural, high-velocity jumps, ensuring that the evolution of the spatial control density obeys physical movement constraints.

## 5.2 Potentials

This section outlines the mathematical formulations for the various potential energy components used in the basketball defensive optimization model. These potentials exert "forces" that collectively guide simulated defenders to optimal defensive configurations.

### 5.2.1 Collective Team-Level Potentials

These potentials are employed to let the defenders collaborate as a whole team.

**Cohesion Potential**: This potential pulls defenders back towards their team's centroid if they stray too far, encouraging defensive formation.

$$E_{\text{cohesion}}(p_d, P_D) = W_{\text{cohesion}} \cdot \max(0, \|p_d - C_D\| - R_{\text{formation}})$$

Where:

- $p_d$: Current position of the defender.
- $P_D$: Set of all defender positions.
- $C_D = \frac{1}{N_D}\sum_{p \in P_D} p$: Centroid of all defender positions.
- $W_{\text{cohesion}}$: Strength of the cohesion pull.
- $R_{\text{formation}}$: The radius within which there is no cohesion penalty.

**Occupancy Penalty**: This potential discourages defenders from occupying the same space, promoting appropriate spacing. It calculates an overlap penalty between pairs of defenders using a simplified circular Gaussian.

For each pair of defenders $(p_i, p_j)$:

- Distance squared: $D_{ij}^2 = \|p_i - p_j\|^2$
- Variance sum: $V_{\text{sum}} = 2 \cdot \sigma_{\text{wide}}^2$
- Overlap term: $O_{ij} = e^{-\frac{D_{ij}^2}{2 \cdot V_{\text{sum}}}}$

The total occupancy penalty is the sum of overlap terms for all unique pairs of defenders:

$$E_{\text{occupancy}}(P_D) = W_{\text{occupancy}} \cdot \sum_{i<j} O_{ij}$$

Where:

- $W_{\text{occupancy}}$: Weight of the occupancy penalty.

**Collective Ball Pressure Potential**: This potential ensures that at least one defender applies pressure to the ball handler. It uses a softmin function to ensure that the minimum energy from any defender to the ball is considered, then blended with a global attraction.

The calculation for each defender involves a blend of local attraction and global distance:

- Local Attraction: $E_{\text{local}}(p_d) = -W_{\text{attraction}} \cdot e^{-\frac{\|p_d - p_{\text{ball}}\|^2}{2\sigma^2_{\text{wide}}}}$
  - $W_{\text{attraction}}$: Weight of local attraction.
- Global Distance: $E_{\text{global}}(p_d) = W_{\text{gb}} \cdot \|p_d - p_{\text{ball}}\|^2$
  - $W_{\text{gb}}$: Weight of global ball attraction.
- Blending Factor: $F_{\text{blend}} = \text{sigmoid}(-k_{\text{blend}} \cdot (\|p_d - p_{\text{ball}}\| - R_{\text{blend}}))$
  - $k_{\text{blend}}$: Blending steepness.
  - $R_{\text{blend}}$: Distance at which blending occurs.

Combined energy for a single defender:

$$E_{\text{single_ball}}(p_d) = (F_{\text{blend}} \cdot E_{\text{local}}(p_d)) + E_{\text{global}}(p_d)$$

The softmin function is then applied to the energies of all defenders with respect to the ball:

$$E_{\text{collective_ball}}(P_D, p_{\text{ball}}) = \text{softmin}(\{E_{\text{single_ball}}(p) \mid p \in P_D\}, k_{\text{softmin}})$$

Where:

- $\text{softmin}(x_1, \ldots, x_N, k) = -\frac{1}{k} \ln\left(\sum_{i=1}^{N} e^{-k \cdot x_i}\right)$
- $k_{\text{softmin}}$: Parameter for the softmin function.

### 5.2.2 Offender Attraction

This potential creates an anisotropic field around each offender, pulling defenders into a specific defensive stance relative to the offender and the basket. The field's strength is dynamically scaled by the offender's threat level.

The calculation involves several steps:

1. **Target Position Calculation**
   Determines an ideal cushion position slightly offset from the offender towards the basket.
   - Vector from offender to basket: $V_{ob} = p_b - p_o$
   - Unit vector: $\hat{V}_{ob} = \frac{V_{ob}}{\|V_{ob}\|}$
   - Target position: $p_{\text{target}} = p_o + D_{\text{cushion}} \cdot \hat{V}_{ob}$
   - Rotation angle: $\theta = \arctan 2(V_{ob,y}, V_{ob,x})$

   Where:
   - $p_o$: Position of the offender.
   - $D_{\text{cushion}}$: Desired distance to stand off the offender.

2. **Rotated Difference** Transforms the defender's position into a coordinate system rotated such that the x-axis aligns with the vector from the target position to the basket.
   - $\Delta x = p_{d,x} - p_{\text{target},x}$
   - $\Delta y = p_{d,y} - p_{\text{target},y}$
   - $x' = \Delta x \cdot \cos(\theta) + \Delta y \cdot \sin(\theta)$
   - $y' = -\Delta x \cdot \sin(\theta) + \Delta y \cdot \cos(\theta)$
3. **Dynamic Threat Scaling**: Adjusts the potential's weight based on the offender's proximity to the ball.
   - Threat scale: $S_{\text{threat}} = 1 + S_{\text{ot}} \cdot e^{-\left(\frac{\|p_o - p_{\text{ball}}\|}{D_{\text{ot}}}\right)^2}$
   - $S_{\text{ot}}$: Scaling factor for offender threat.
   - $D_{\text{ot}}$: Distance threshold for offender threat.
   - $p_{\text{ball}}$: Position of the ball.
4. **Anisotropic Potential**:

$$E_{\text{field}}(p_d, p_o, p_{\text{ball}}) = W_{\text{field}} \cdot S_{\text{threat}} \cdot e^{-\left(\frac{(x')^2}{2\sigma_{\text{long}}^2} + \frac{(y')^2}{2\sigma_{\text{wide}}^2}\right)}$$

   Where:
   - $W_{\text{field}}$: Base weight of the anisotropic field (negative for attraction).
   - $\sigma_{\text{long}}$: Standard deviation along the hoop-facing axis.
   - $\sigma_{\text{wide}}$: Standard deviation along the sideways axis.

The total $E_{\text{field}}$ is the sum of potentials from all offenders.

### 5.2.3 Basket Anchor

This potential attracts defenders towards the basket, acting as a baseline defensive position. It's an exponential function, creating a strong pull as the defender gets further away from the basket.

$$E_{\text{gravity}}(p_d) = -W_{\text{bg}} \cdot e^{-\frac{\|p_d - p_b\|^2}{2\sigma_{\text{bg}}^2}}$$

Where:

- $p_d$: Current position of the defender.
- $p_b$: Position of the basket.
- $W_{\text{bg}}$: Weight of the basket gravity potential.
- $\sigma_{\text{bg}}$: Standard deviation, controlling the range/spread of the basket's influence.

### 5.2.4 Boundary Penalty Potential

This potential penalizes defenders for getting too close to the court boundaries, ensuring they stay within playable areas.

For each defender $p_d = (x, y)$:

$$E_{\text{boundary}}(p_d) = W_{\text{boundary}} \cdot [\max(0, B-x)^2 + \max(0, B-(X_{\text{court}}-x))^2 + \max(0, B-y)^2 + \max(0, B-(Y_{\text{court}}-y))^2]$$

Where:

- $B$ (2.0): Buffer distance from the boundary.
- $X_{\text{court}}$ (94): Court width.
- $Y_{\text{court}}$ (50): Court height.
- $W_{\text{boundary}}$ (10.0): Weight of the boundary penalty.

### 5.2.5 IST Potential

While the previous potentials establish physical constraints, they treat all offensive players potentials equally. To bridge the gap between kinematics and basketball strategy, we integrated the Instantaneous Shot Threat (IST) formulated in section 4.3 directly into the energy functional.

This potential acts as a specialized spatial penalty. By minimizing this total energy, the defense ($P_D$) is incentivized to reduce the "openness" of the most dangerous offensive players on the floor.

$$E_{\text{IST}}(P_D, P_O, p_{\text{ball}}) = W_{\text{ist}} \cdot \sum_{o \in P_O} \text{IST}(p_o)$$

Where:

- $W_{\text{ist}}$: Scaling weight for the threat-based attraction.
- $\text{IST}(p_o)$: Individual player threat as defined in Eq. 2.
- $k_{\text{smooth}}$: Smoothing constant for the internal `softmin` operator (defined in 4.3).

By coupling this data-driven penalty with physical JKO constraints, defenders are able to prioritize threats dynamically, mimicking the high-level decision-making found in professional play.

## 5.3 Bayesian Optimization

Bayesian Optimization (BO) is a strategy for finding the global optimum of a complicated black-box function. This section explains the mathematical formulations for the loss metrics we used to evaluate defensive strategies. These metrics are used by Optuna from Akiba et al. (2019) for hyperparameter optimization of potentials in the previous section.

### 5.3.1 Loss Functions

To evaluate performance of a parameter set, we defined three different loss calculated across the simulated defensive trajectories:

**1. Pressure Loss ($L_{\text{pressure}}$)**

Pressure loss component quantifies how effectively defenders are guarding offenders, with a bias towards being on the goal side of the offender. A lower value indicates better pressure.

1. Standard Distance Matrix
   This calculates the Euclidean distance between every offender and every defender. Let $P_O = \{p_{o,1}, \ldots, p_{o,N_o}\}$ be the set of offender positions and $P_D = \{p_{d,1}, \ldots, p_{d,N_d}\}$ be the set of defender positions.
   The distance matrix $D$ has entries:

$$D_{ij} = \|p_{o,i} - p_{d,j}\|$$

2. Goal-Side Factor
   This factor biases the pressure metric to prefer defenders being between the offender and the basket.
   (a) Vector from Offender to Basket
   For each offender $i$,

$$V_{oob,i} = p_b - p_{o,i}$$

   (b) Unit Vector from Offender to Basket

$$\hat{V}_{oob,i} = \frac{V_{oob,i}}{\|V_{oob,i}\|}$$

   (c) Vector from Offender to Defender
   For each offender $i$ and defender $j$,

$$V_{ood,ij} = p_{d,j} - p_{o,i}$$

   (d) Unit Vector from Offender to Defender

$$\hat{V}_{ood,ij} = \frac{V_{ood,ij}}{\|V_{ood,ij}\|}$$

   (e) Alignment
   The dot product measures how aligned the defender is with the basket relative to the offender.

$$A_{ij} = \hat{V}_{oob,i} \cdot \hat{V}_{ood,ij}$$

   An $A_{ij}$ of 1 means the defender is directly between the offender and the basket. An $A_{ij}$ of -1 means the defender is behind the offender (from the basket's perspective).
3. Positional Penalty
   This penalty scales the distance based on the alignment. Defenders behind the offender receive a higher penalty.

$$P_{pos,ij} = 1.5 - 0.5 \cdot A_{ij}$$

$$\text{This ensures that } \begin{cases} P_{pos,ij} = 1.0, & \text{if } A_{ij} = 1 \text{ (goal-side)} \\ P_{pos,ij} = 2.0, & \text{if } A_{ij} = -1 \text{ (behind)} \end{cases}$$

4. Weighted Distance
   The standard distance is multiplied by the positional penalty:

$$D'_{ij} = D_{ij} \cdot P_{pos,ij}$$

   For each offender $i$, we take the minimum weighted distance to any defender:

$$\min_D'_i = \min_j(D'_{ij})$$

5. Danger Weight
   This weights offenders based on their proximity to the basket, giving more importance to offenders closer to the hoop.

$$W_{danger,i} = \text{softplus}(30 - \|p_{o,i} - p_b\|)$$

   Where $\text{softplus}(x) = log(1 + exp(x))$ which is a smooth approximation of $\max(0, x)$. This creates a higher weight for offenders closer to the 30-foot mark from the basket.
6. Final Pressure Calculation The average of the minimum weighted distances, scaled by their danger weights:

$$L_{pressure,t} = \frac{1}{N_o} \sum_{i=1}^{N_o} (\min_D'_i \cdot W_{danger,i})$$

**2. Threat Minimization Loss ($L_{\text{IST}}$)**

While pressure loss measures raw geometry, the IST loss measures success by evaluating the defenses ability to suppress Instantaneous Shot Threat (defined in Section 4.3) of all 5 offenders:

$$L_{\text{IST}} = \frac{1}{T} \sum_{t=1}^{T} \sum_{p_o \in P_O} \text{IST}\left(p_o, P_D^{(t)}\right)$$

**3. Smoothness Loss ($L_{\text{smoothness}}$)**

This metric penalizes jerky or abrupt movements in the defensive trajectory, promoting more fluid and realistic player movement. It's calculated as the mean Euclidean distance of the change in position between consecutive frames for all defenders.

Let $P_{D,t}$ be the set of defender positions at time $t$.

$$L_{smoothness} = \frac{1}{(T-1) \cdot N_d} \sum_{t=0}^{T-2} \sum_{j=1}^{N_d} \|p_{d,j,t+1} - p_{d,j,t}\|$$

Where $T$ is the number of timesteps and $N_d$ is the number of defenders.

The overall pressure loss and smoothness loss are the mean of these values across all timestep in the simulation. These two objectives are minimized simultaneously by Optuna during hyperparameter tuning.

### 5.3.2 Two-Stage Optimization

Because the integration of the IST potential introduces highly non-linear gradients, optimizing all physics parameters simultaneously casued the objective function to become unstable. To resolve this, we used a Two-Stage Bayesian Optimization strategy through the Optuna framework.

1. **Stage 1 (Kinematic Baseline)**: We first optimized foundational geometric variables (cohesion, occupancy, boundary, and ball pressure) against Pressure Loss ($L_{\text{pressure}}$) and Smoothness Loss ($L_{\text{smoothness}}$). This established a kinematic baseline defense that obeyed physical laws, but lacked strategic prioritization.
2. **Stage 2 (Threat Integration)**: We then froze the optimal geometric parameters from Stage 1 and isolated the IST variables. We ran a secondary optimization, targeting the Threat Minimization Loss ($L_{\text{IST}}$) and $L_{\text{smoothness}}$ to tune the influence of offender weights.

This staged coordinate descent converged smoothly, yielding optimal threat parameters of $W_{\text{ist}} = 1.479$ and $k_{\text{smooth}} = 9.427$. By isolating the parameter spaces, this approach ensured that the integration of player shot threat improved, rather than interfered with the Wasserstein gradient flow.

## 5.4 Solver

This section details the mathematical and algorithmic components of the JKO solver which is responsible for simulating defensive player movements using JAX from Bradbury et al. (2018) and Optimal Transport concepts.

### 5.4.1 Loss Function

The JKO solver steps the simulation forward by minimizing a composite loss function to find the optimal defender positions ($P_{D,\text{cand}}$), given their previous ($P_{D,\text{prev}}$) and previous-previous ($P_{D,\text{prev_prev}}$) states.

Let $P_O$ be offender positions, $p_b$ be ball position, and $P_{\text{basket}}$ be basket position.

1. **Potential Energy**
   This term represents the defensive strategy's goal and is computed by summing all of the potentials for all defenders, along with the global Instantaneous Shot Threat (IST) penalty.

$$L_{\text{potential}} = \left( \sum_{p_d \in P_{D,\text{cand}}} E(p_d, P_{D,\text{cand}}, P_O, p_b, P_{\text{basket}}, \text{params}) \right) + E_{\text{IST}}(P_{D,\text{cand}}, P_O, p_b)$$

2. **Kinetic Energy**
   This term quantifies the cost of moving defenders from $P_{D,\text{prev}}$ to $P_{D,\text{cand}}$ using the

Sinkhorn divergence, scaled by JKO lambda coefficient. This encourages smooth, efficient movements. Sinkhorn divergence will be addressed in the next section.

$$L_{\text{kinetic}} = \frac{1}{\lambda_{\text{jko}}} \cdot W_2(P_{D,\text{cand}}, P_{D,\text{prev}}, \varepsilon_{\text{sinkhorn}})$$

Where $W_2$ is the Wasserstein-2 distance, and $\lambda_{\text{jko}}$ and $\varepsilon_{\text{sinkhorn}}$ are parameters.

3. **Acceleration Penalty**
   This penalty discourages rapid changes in velocity, making movements more fluid and less vibrating. It is based on the discrete approximation of acceleration.
   For each defender $j$:

$$\text{accel}_j = p_{d,\text{cand},j} - 2 \cdot p_{d,\text{prev},j} + p_{d,\text{prev_prev},j}$$

   The penalty is:

$$L_{\text{acceleration}} = W_{\text{accel}} \cdot \sum_j \|\text{accel}_j\|^2$$

   Where $W_{\text{accel}}$ is the weight of this penalty.

4. **Velocity Penalty**
   This penalty discourages high speeds, even if they are within the velocity cap, acting as a "friction" term.

$$L_{\text{velocity}} = W_{\text{vel}} \cdot \sum_j \|p_{d,\text{cand},j} - p_{d,\text{prev},j}\|^2$$

   Where $W_{\text{vel}}$ is the weight of this penalty.

The total loss is the sum of all of these losses:

$$L_{\text{total}} = L_{\text{potential}} + L_{\text{kinetic}} + L_{\text{acceleration}} + L_{\text{velocity}}$$

### 5.4.2 Sinkhorn Divergence

Sinkhorn divergence, a differentiable approximation of the Wasserstein-2 distance, is used to quantify the cost of moving a distribution of defenders from one configuration to another. It measures the squared cost of optimally transporting defenders from their previous positions to their current proposed positions.

The formula for Sinkhorn divergence $S_\varepsilon(X, Y)$ between two sets of points $X = \{x_i\}_{i=1}^N$ and $Y = \{y_j\}_{j=1}^N$ with entropy regularization $\varepsilon$ is:

$$S_\varepsilon(X, Y) = \min_{\Pi \geq 0, \sum_j \Pi_{ij} = \frac{1}{N}, \sum_i \Pi_{ij} = \frac{1}{N}} \left( \sum_{i,j} C_{ij} \Pi_{ij} + \varepsilon \sum_{i,j} \Pi_{ij} \log \Pi_{ij} \right)$$

Where $C_{ij} = \|x_i - y_j\|^2$ is the cost matrix (squared Euclidean distance).

## 5.5 Simulation Pipeline

To evaluate the efficacy of the JKO scheme, we engineered a large-scale simulation pipeline capable of processing thousands of continuous NBA possessions.

For each possession (complied using the trajectory alignment from in Section 3) the pipeline initializes all ten players at the start of the temporal window. While the offensive players follow their real trajectories, the defensive players are controlled by the JKO physics engine (with first frame positioning in the temporal window). The engine calculates the optimal movement for each defender at every frame, allowing the defense to dynamically react to the offensive play.

We ran this batch simulation across the validation and test sets, requiring about 6 hours of compute time. To compare results, the pipeline exports three key outputs per possession:

- **JKO Defensive Trajectorie**: The coordinate matrices ($P_{D,\text{sim}}$) generated by the JKO solver.
- **Real-World Tracking Data**: The coordinate matrices ($P_{D,\text{real}}$) of the real-world NBA defenders .
- **Delta Threat Mectrics**: The frame-by-frame Instantaneous Shot Threat (IST) for both the simulated and real defenses.

This unified dataset allows us to statistically analyze whether the our model limits scoring threats more effectively than real NBA defenders.

# 6 Results

This section presents the empirical findings from the simulation pipeline, comparing the performance of JKO-simulated defenders against real NBA tracking data.

## 6.1 Metric Validation

To establish the baseline validity of our Instantaneous Shot Threat metric, IST values of the real defense was compared against field goal success and team-level defensive ratings.

As shown in Figure 3, actual NBA Field Goal percentages (FG%) vary across the five defined pressure tiers. Shots categorized as “Wide Open” (Tier 1) yielded a 48.46% success rate, while “Smothered” (Tier 5) shots dropped to 42.3%, a 12.7% decrease.

At the team level, Figure 4 illustrates the relationship between a team’s Average Real IST per play and their official 2015-16 NBA Defensive Rating. The data indicates a correlation of $r = 0.307$, an $R^2 = 0.094$, and a $p$-value of $9.9 \times 10^{-2}$.

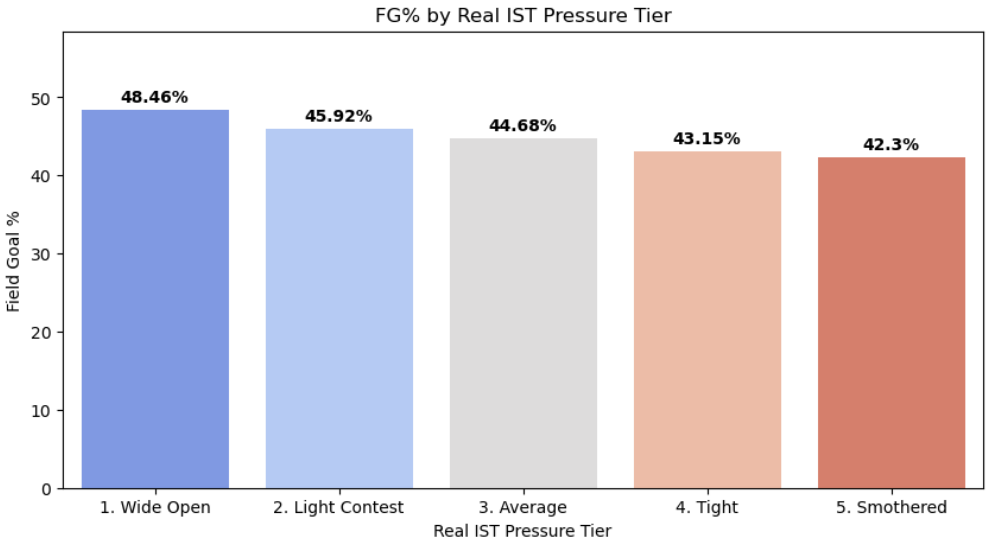



Figure 3: **Field Goal Percentage by Real IST Pressure Tier.** Historical FG% for all shot events in the dataset categorized into five discrete pressure tiers. Tier 1 represents the lowest topological pressure, Tier 5 represents the highest.

## 6.2 Aggregate Threat Reduction

The primary objective of the JKO scheme is to minimize the offensive team's spatial scoring probability (Lower IST = better). Per play, real NBA defenders recorded a mean Team IST ($\mu_{\text{real}}$) of 238.73, while the JKO-simulated defenders ($\mu_{\text{sim}}$) recorded a mean of 157.33.

As shown in Figure 5, this represents a 51.73% reduction in mean threat. The JKO physics engine achieved a global aggregate Frame Win Rate of 97.12%, which represents the total percentage of simulation frames where the simulated defense yielded a lower Team IST than the real NBA defense.

## 6.3 Defensive Recovery in High-Threat Scenarios

To evaluate the models effectiveness in critical defensive breakdowns, a "high-threat" subset was isolated. This subset consists of possessions where the real-world NBA defense allowed a total threat in the 90th percentile of all historical plays, establishing a high-threat threshold of 298.16 IST units.

- **Median Recovery Gap**: Within this high-threat subset, the median raw difference between the historical Real IST and the JKO Simulated IST was 125.75 units, , representing a median threat reduction of 39.39%..
- **High-Threat Play**: As shown in Figure 6, this gap identifies the scoring probability reduction by the JKO engine, the simulation consistently maintained a lower IST threat level throughout the high-threat play.

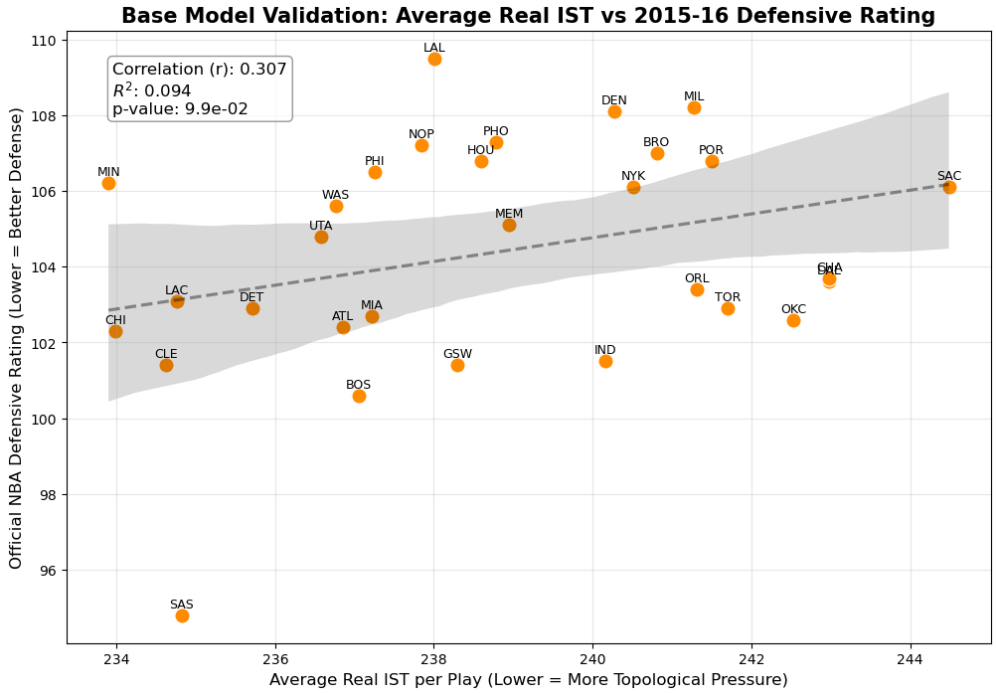



Figure 4: **Mean Real IST vs. NBA Defensive Rating (2015-16).** Correlation between topological pressure and official defensive efficiency across all 30 NBA teams. The dashed line represents the linear regression, and the gray region indicates the 95% confidence interval.

# 7 Discussion

The results of the JKO defensive simulation demonstrate a significant difference in defensive scheme compared to real NBA defense. By prioritizing mathematical consistency over human intuition, the model gives us new insights on how we could "optimally" play defense.

## 7.1 Interpreting System Reliability

The correlation between Real IST and Field Goal Percentage (Figure 3) confirms that defensive pressure is a somewhat reliable metric for rating defense. The 6.16 percentage point drop in FG% from "Wide Open" to "Smothered" tiers suggests that the IST potential function successfully captures the physical constraints of an effective contest.

The team-level correlation ($r = 0.307$) with official Defensive Ratings (Figure 4) provides a macro-validation for the model. Given the fixed sample of $N = 30$ NBA teams, even with a p-value of 0.099, this result is statistically meaningful. Our model being able to explain nearly 10% of defensive variance highlights the importance of positioning when playing defense.

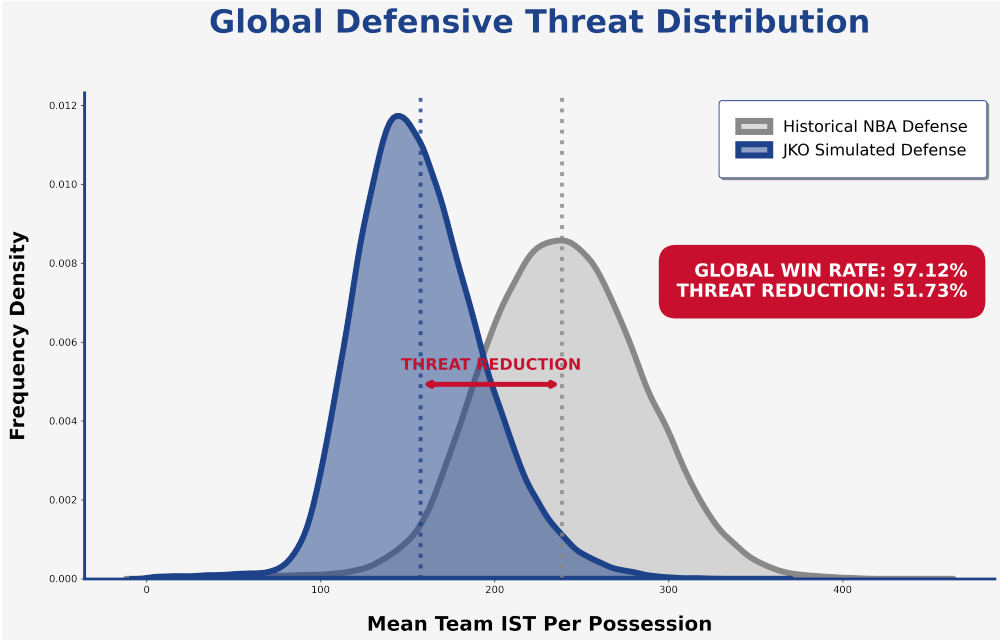



Figure 5: **Global Distribution of Team IST and System Reliability.** This Kernel Density Estimate (KDE) plot compares the total offensive threat allowed per possession. The vertical dashed lines denote distribution means ($\mu_{\text{real}} = 238.73, \mu_{\text{sim}} = 157.33$). The dashboard in the upper-right corner displays the combined 97.12% Global Win Rate and the 51.73% Mean Threat Reduction.

## 7.2 Mathematical Consistency

One of the most compelling results we found is the is the 97.12% Global Frame Win Rate. This indicates that in nearly every micro-movement of a possession, there is a gradient-flow solution that is mathematically safer when compared to real athletes.

The high-threat play analysis gave a Median Recovery Gap of 125.75 units and a median threat reduction of 39.39%. As recorded in the temporal audit (Figure 6), the historical baseline exhibits localized threat "spikes" during specific offensive actions, this could be due to a variety of factors, such as delayed recognition, fatigue, or bad defensive matchups.

## 7.3 Limitations/Future Works

While the JKO simulation demonstrates improvements in defensive positioning, several physical and structural constraints remain for future optimization.

### 7.3.1 Speed Constraints

Ideally, our simulated defenders should move around the actual defender speed range. However, due to several reasons, Simulated defenders moves more than twice than the

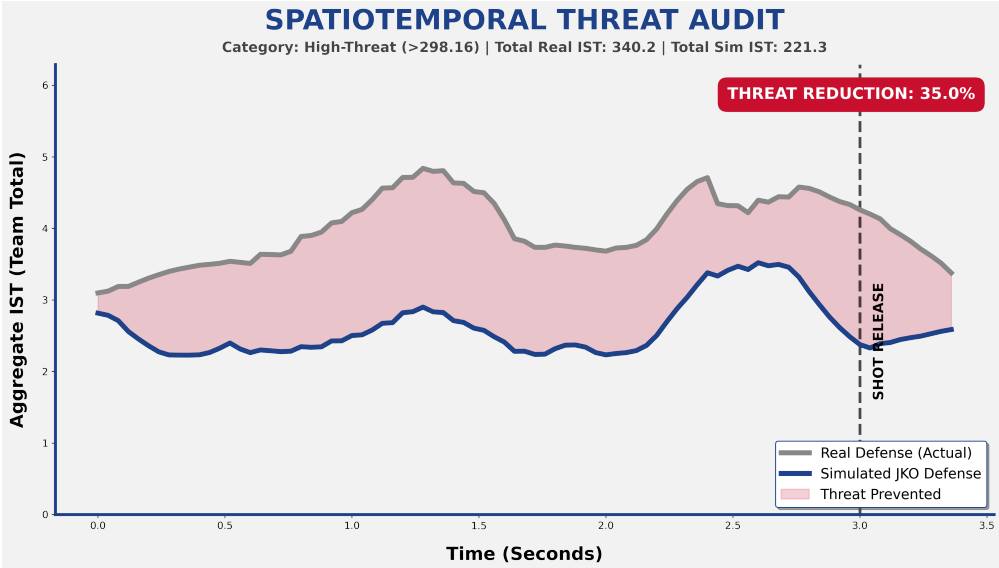



Figure 6: **High-Threat Recovery Audit.** A temporal comparison of aggregate Team IST for a possession exceeding the 298.16 threshold. Metadata included in the chart shows the IST threat levels allowed by the historical baseline versus the JKO optimized response

real defenders even with the speed constraints, shown in Figure 7.

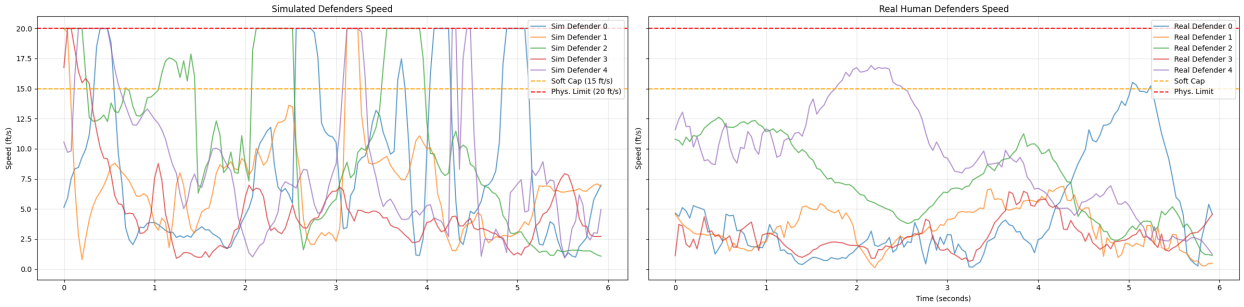



Figure 7: Speed comparison between the real defenders and simulated defenders

We tried several methods to restrict the movement of the players. But all of them either

1. failed to move the players fast enough to escape a certain gravity well, especially the ones created by other defenders.
2. could not control the speed in non-necessary situations, and made the players always move at the soft-cap or above.

Our current implementation were able to take the middle point of these problem and is in the middle state that defenders occasionally surpasses the soft speed limit and reaches the hard limit.

Although the speed looks too fast than the real defenders, this does not deny our simulation success. One of the most important reason this phenomenon happens is due to the steep Gaussian. Although JKO scheme is able to control this at a certain level, it sill fails to

manage at some cases. Figure 7 also indicates it as the speed overpasses the soft limit only in a short instant. This suggests that the defenders were able to stop right after the acceleration, which means that they reach the bottom of the gravity well.

However, there are still more difficulties around speed. First of all, It is hard to say that the result is great while we cannot solve the speed constraints problem. While some simulation looks like a very natural defense to a human eye, some simulation fails to pass even the eye test due to the lack of velocity control. The problem here is that we cannot tell what exactly the cause of the improve in defensive evaluation; if it was simply our simulation working better than the real defenders or our simulated defenders were able to move much more than an NBA player could not do realistically.

### 7.3.2 Offender Threat Potential

While we have successfully implemented a Offensive Potential to pull defenders and create a defense formation, there are still a lot of rooms to improve. The biggest problem here is that the defenders sees the off-the-ball offensive threat through only the distance to the ball. For future extension, it is ideal that the offensive potential changes based on their momentum and even more, their posture using Graph Convolution Neural Network.

### 7.3.3 Heterogeneous Defensive Profiles

Our model assumes defensive homogeneity, treating all five defenders as mathematically identical agents. This isn't the case in the real NBA, where defensive schemes can be centered around unique player archetypes, such as elite interior rim-protectors like Victor Wembanyama, or high-pressure perimeter defenders like Alex Caruso. Future iterations could incorporate individual defense impact metrics such as wingspan, lateral quickness, or defensive win-shares.

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